

MLFF campaign storage-management specification (Transitional P6/P7 Contract)

Status: Accepted transitional specification under P6/P7 governance. Historical STOR1-STOR5 design notes are archived in docs/history/mlff/STOR1_STOR5_HISTORICAL_DESIGN.md. Consequential storage transformations are deferred to the unified storage reset workplan CODE-MLFF-CAMPAIGN-STORAGE-IO-RESET1.

Purpose

This specification defines the authoritative transitional storage-management contract for MLFF training data campaigns during the P6/P7 lifecycle. Storage policy is strictly subordinate to scientific provenance, training restartability, and user data ownership.

Non-negotiable protections

1. **External directory immutability:** User-supplied source datasets, training sets, replay trajectories, and true-label reference material located outside the campaign workspace are read-only inputs. Campaign storage operations shall never delete, truncate, rewrite, rename, or relocate external paths.
2. **Ownership and boundary enforcement:** A path reference appearing in TOML configuration, SQLite state, JSON metadata, or an artifact manifest does not confer deletion authority. Destructive operations are strictly restricted to campaign-owned artifacts that pass physical containment and ownership boundary validation.
3. **Symlink traversal protection:** A symlink object located inside the campaign workspace may be unlinked only if campaign-owned, but cleanup shall never traverse external symlink targets.
4. **Production artifact preservation:** Final selected production models, production checkpoints, and selection/verification evidence are retained by default.
5. **Fail-toward-retention under incomplete lifecycle:** In the presence of incomplete stages, active training jobs, unverified evidence, or ambiguous ownership boundaries, cleanup fails closed toward full artifact retention.

1. Storage Accounting (storage report)

The `storage report` command is strictly read-only. It inspects the campaign workspace and produces `results/storage-report.json` reporting: - Logical allocated bytes and unique-inode bytes; - File counts and artifact family distributions; - Protected input boundaries and external reference validations; - Reconstructable cache sizes versus retained scientific evidence.

No files are modified or removed during accounting operations.

2. Safe and Cache Cleanup (storage cleanup)

Current-generation campaigns support two non-destructive / low-consequence cleanup tiers:

Default / Safe Tier (`--tier safe`)

- Targets only temporary scratch, aborted stage staging trees, obsolete runtime scratch after compact diagnostics, and orphaned external database records.
- Guarantees zero loss of scientific capability, evaluation state, or training restartability.
- Every safe cleanup event appends an authenticated record to `results/cleanup-manifest.jsonl` with an empty capability-loss set.

Cache Tier (--tier cache)

- Targets independently reconstructible acceleration caches (e.g. normalized frame-cache and non-authoritative checkpoint-model-cache).
 - Removal of cache files incurs no scientific loss, but future stage execution may re-derive normalized frames from protected inputs.
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3. Quarantined Operations Deferred to Post-P7 Storage Reset

Consequential historical storage tiers and physical transformations change reanalysis, continuation, or file-structure semantics and are **not supported** for current P6/P7 generation campaigns: `- storage cleanup --tier recompute - storage cleanup --tier compact - storage cleanup --tier archive - storage deduplicate --apply - storage archive create - storage archive restore`

When invoked on a current-generation campaign, these commands fail closed with a clear `CampaignCliError` indicating that consequential storage transformation is deferred to `CODE-MLFF-CAMPAIGN-STORAGE-IO-RESET1`.